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Digio Click (DJOClick)
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BUSINESS REQUIREMENTS DOCUMENT (BRD)

UrjaMitra AI (ఉర్జామిత్ర)

Agentic Personalized Billing & Consumer Copilot for TGSPDCL & TGNPDCL

INITIATIVE	TARGET DISCOMS	PROBLEM STATEMENT ID	DATE & STATUS
AIKAM EnergyTech Challenge	**TGSPDCL & TGNPDCL**	**885fa752 (Billing Assistance)**	**October 2026 | Approved Baseline**

1. EXECUTIVE SUMMARY & PROGRAM CONTEXT

The Government of Telangana, in collaboration with AIKAM and the Department of Energy, has launched the EnergyTech Innovation Challenge to deploy transformative artificial intelligence across the state's electricity distribution utilities: **TGSPDCL** (Telangana Southern Power Distribution Company Limited) and **TGNPDCL** (Telangana Northern Power Distribution Company Limited).

Across more than **1.75 Crore (17.5 Million)** registered electricity connections in Telangana, billing inquiries, tariff slab confusion, Gruha Jyothi subsidy status questions, and meter reading disputes account for over **68% of all inbound citizen calls** to the 1912 utility hotline and section customer care offices.

> **The UrjaMitra AI Mission**
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> **UrjaMitra AI** is an autonomous, multi-agent conversational and billing copilot engineered specifically for Telangana power utilities. It bridges the communication gap between complex regulatory tariff orders (TGERC slabs, Gruha Jyothi zero-bill eligibility, fuel surcharge adjustments) and citizens across urban, peri-urban, and rural Telangana via **WhatsApp, Web, and Voice in Telugu, Hindi, and English**.

2. DEEP TOPIC INDEXING MATRIX

Topic ID	Requirement Area	Scope Category	Stakeholder Lead	Target Phase
BRD-01	Executive Problem & Tariff Friction	Core Business Context	Solutions Lead (M. Prabhu)	Phase 1 (Immersion)
BRD-02	Stakeholder Personas & Journey Maps	User Experience & Ops	Delivery Lead	Phase 1 (Immersion)
BRD-03	Quantitative Business Value & ROI	Utility FinOps	Solutions Lead	Phase 1 (Immersion)
BRD-04	Functional Scope (Agents 1 through 4)	Agentic Product Engine	Product Architect	Phase 2 (Oct 11)
BRD-05	Regulatory & TGERC Compliance	Governance & Legal	Domain Expert	Phase 2 (Oct 11)
BRD-06	Integration with DISCOM SAP Systems	Enterprise Architecture	SAP Systems Lead	Phase 3 (Oct 14)

Topic ID	Requirement Area	Scope Category	Stakeholder Lead	Target Phase
BRD-07	Non-Functional Specs & DPDP Act 2023	Security & Reliability	DevSecOps Lead	Phase 3 (Oct 14)
BRD-08	Pilot Rollout (Circles & Feeders)	Operational Rollout	Program Manager	Phase 4 (Oct 25 Demo)

3. PROBLEM STATEMENT & ROOT CAUSE ANALYSIS

1. Telescopic Slab Transitions & Citizen Bill Shock

Telangana domestic LT Category-I billing operates under telescopic slab rates with consumption brackets: Band A (0–100 kWh), Band B (101–200 kWh), and Band C (>200 kWh). Crossing even 1 single unit from Band B (199 units) to Band C (201 units) transitions the entire base energy calculation into higher per-unit rates, accompanied by sharp escalation in monthly customer charges. Citizens perceive this as an arbitrary meter jump.

2. Gruha Jyothi 200-Unit Cliff Edge

Under the flagship **Gruha Jyothi Scheme**, eligible domestic consumers consuming ≤ 200 units receive a **Zero Bill** (100% state subsidy). If consumption hits 201 units, the subsidy is nullified or significantly curtailed under government rules, triggering a sudden ₹1,000+ bill. Citizens flood MeeSeva centers and DISCOM offices seeking answers.

3. Door Lock (Status 02) & Average Billing Accumulations

When physical meter readers encounter locked gates or damaged optical sensors, the utility billing engine records "Average Billing" (Status 02). In subsequent months, when an actual meter reading is performed, the accumulated multi-month consumption pushes the consumer into the highest tariff tier (₹10–11/unit), creating severe dispute escalations.

4. STAKEHOLDER PERSONAS & INTERACTION MODELS

Persona	Role	Primary Pain Point	UrjaMitra AI Value Proposition
Citizen (Urban / Rural)	Electricity Consumer (LT-I / LT-II)	Confused by slab calculations, speaks Telugu/Hindi, fears meter tampering fraud.	Instant Telugu voice/WhatsApp explanation, line-by-line slab breakdown, advice on staying under 200 units.
1912 Call Center Agent	DISCOM Customer Support	Handles 150+ repetitive calls daily with 8-minute handle times; high burnout.	Deflects 65% of routine queries autonomously; receives pre-validated dispute tickets for the rest.
Section Officer (AAE / ADE)	Field Operations Engineer	Overwhelmed by paper dispute files and meter re-reading requests.	Receives OCR meter verification packages with photo proof directly in SAP IS-U queue.
CGM (Commercial / Revenue)	TG-DISCOM Revenue Head	Delayed collections, escrow reconciliations, high cost-to-serve, consumer distrust.	Boosts collection efficiency by 4.2%, slashes dispute cycle from 18 days to 2 hours.

5. QUANTITATIVE BUSINESS CASE & ROI METRICS

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Annual Utility Savings = Savings(Call Center) + Savings(Field Visits) + Savings(Early Liquidity) - Opex(UrjaMitra)
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1. Call Center Cost Reduction (1912 Hotline):

TGSPDCL and TGNPDCL receive approximately 420,000 billing inquiries monthly across 1912 and circle offices. Average cost per human resolution: **₹14.50**. With UrjaMitra AI deflecting 65% (273,000 queries deflected), the monthly savings equal: **273,000 × ₹14.50 = ₹39.58 Lakhs / month (≈ ₹4.75 Crores annually)**.

2. Field Re-reading & Dispute Visit Elimination:

18,000 monthly field inspection tickets dispatched for disputed meter readings (costing ₹85 per engineer visit). With Multimodal Meter OCR resolving 70% of photo disputes instantly, the utility saves:

12,600 × ₹85 = ₹10.71 Lakhs / month (≈ ₹1.28 Crores annually).

3. Working Capital & Cash Realization:

Bill clarity and transparent True-Up explanations reduce dispute holding periods by 12 days, accelerating utility operating liquidity by **₹18+ Crores**.

6. FUNCTIONAL REQUIREMENTS BREAKDOWN

FR-1: Telescopic Slab Deconstruction Engine

Ingests Unique Service Number (USC No) or bill amount and itemizes Energy Charges per bracket (0–100, 101–200, 201–300, 301–400, >400 units), Fixed Charges, Customer Charges, and Electricity Duty. For Gruha Jyothi consumers, clearly highlights zero-bill subsidy grant and exact units remaining before crossing the 200-unit threshold.

FR-2: Multimodal Bill & Meter Vision OCR

Accepts physical paper bill photos, PDFs, and live meter display photos. Extracts USC No, Billing Date, Previous Reading, Present Reading, and Status (01 Normal, 02 Door Lock). Automatically compares live consumer meter photos against billed values to identify Average Billing errors.

FR-3: Multilingual Vernacular Voice & Chat Copilot

Native support in **Telugu (తెలుగు)**, **Hindi (हिन्दी)**, and **English**. Handles regional accents and mixed Telugu-English phrasing. Delivers conversational responses via Web, WhatsApp Business Cloud API, and 1912 IVR voice bridge.

FR-4: Proactive Consumption & Spike Advisory

Clusters consumption into Time-of-Day (ToD) buckets and alerts consumers before they breach subsidy slabs (e.g., "*You used 188 units with 4 days remaining. Staying under 200 units saves ₹1,200 under Gruha Jyothi*"). Includes PM Surya Ghar rooftop solar ROI calculations.

FR-5: Automated Dispute Resolution & SAP Ticket Agent

Calculates provisional revised bills for verified anomalies, packages cryptographic evidence (image hash, OCR confidence, timestamp), and posts directly into the TGSPDCL/TGNPDCL SAP IS-U complaint queue with human-in-the-loop Section Officer approval.

7. REGULATORY COMPLIANCE & DPDP ACT 2023

TGERC Retail Supply Tariff Alignment: All calculation engines are bound strictly to the latest Telangana State Electricity Regulatory Commission (TGERC) Gazette notifications. No LLM hallucinations are permitted in monetary amounts.

Digital Personal Data Protection (DPDP) Act 2023: Full adherence to citizen privacy laws. Consumer identifiers, phone numbers, and bank details are encrypted using AES-256 at rest and masked in analytical logs. Compute and storage reside exclusively in MeitY-empanelled data centers in India.

8. KEY PERFORMANCE INDICATORS (KPIS)

Metric Name	Current Baseline	Target with UrjaMitra AI	Full State Target
First-Contact Resolution (FCR)	32%	**78%**	88%
Average Dispute Resolution Cycle	14 to 21 Days	**< 4 Hours**	< 2 Minutes
Vernacular Adoption (Telugu/Hindi)	12% (Web)	**> 70% (WhatsApp/Voice)**	> 85%
1912 Hotline Deflection Rate	0%	**65% Deflection**	75% Deflection

Metric Name	Current Baseline	Target with UrjaMitra AI	Full State Target
Consumer CSAT Score	2.1 / 5.0	**4.3 / 5.0**	4.6 / 5.0